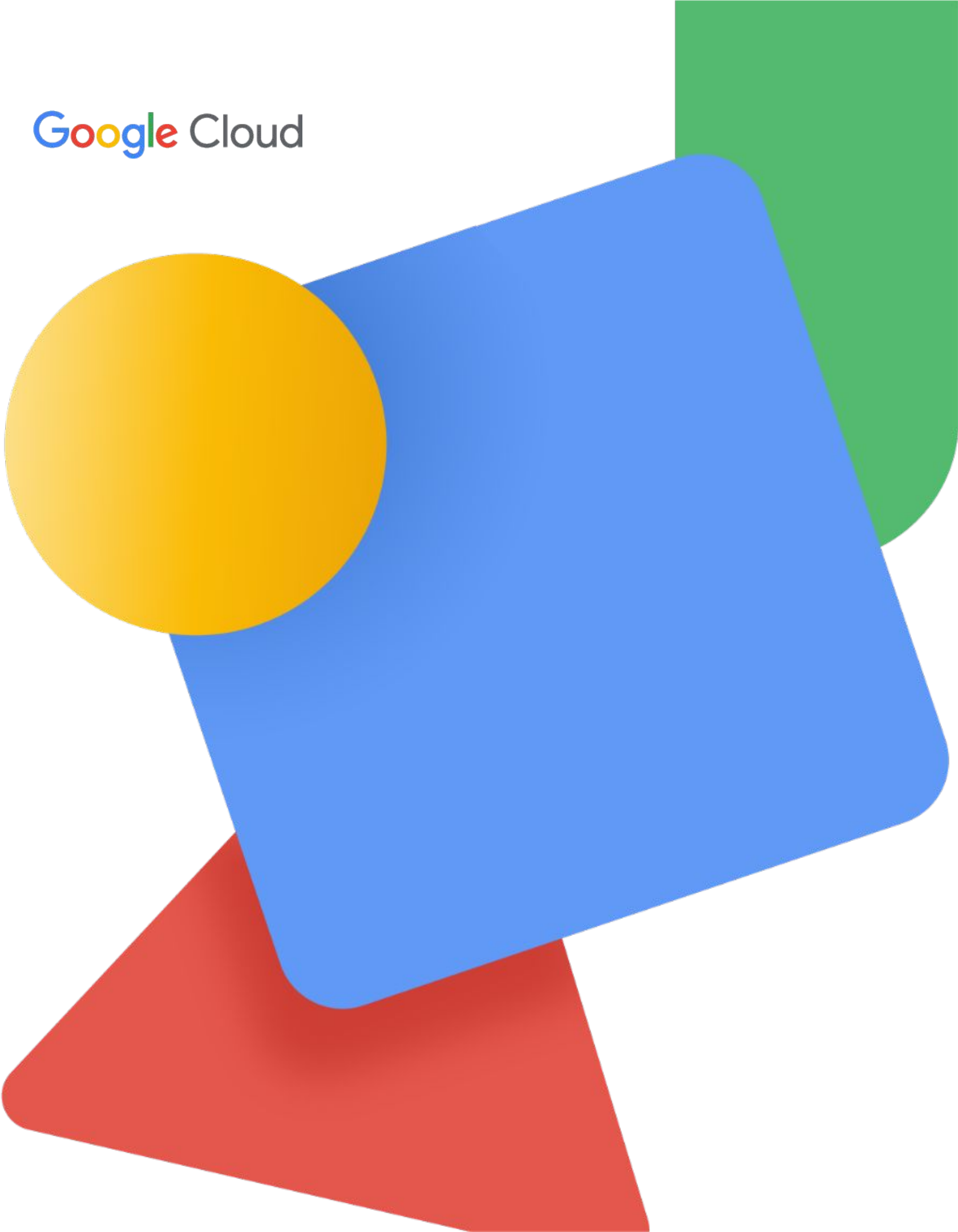


Welcome — who this course is for

Iowa DOM DoIT data team · Oct 5–6, 2026 · 9:00 AM–4:30 PM CT

- ✓ ETL developers living in SQL Server and SSIS— sources, transforms, sinks are your daily work
- ✓ A dedicated M&O team— you will own what gets designed here, so operations questions are welcome
- ✓ Power BI report authors— the last mile matters: your Gold marts feed the reports you already have
- ✓ Near-zero GCP console experience across the room— that is expected, and this course is built for it
- ✓ Senior in data engineering, new to one console— we compress concepts, slow down on where things live

05:00

Abstract geometric shapes in blue, green, yellow, and red on the left side of the slide.

Introduction to Data Engineering on Google Cloud

Name
Role

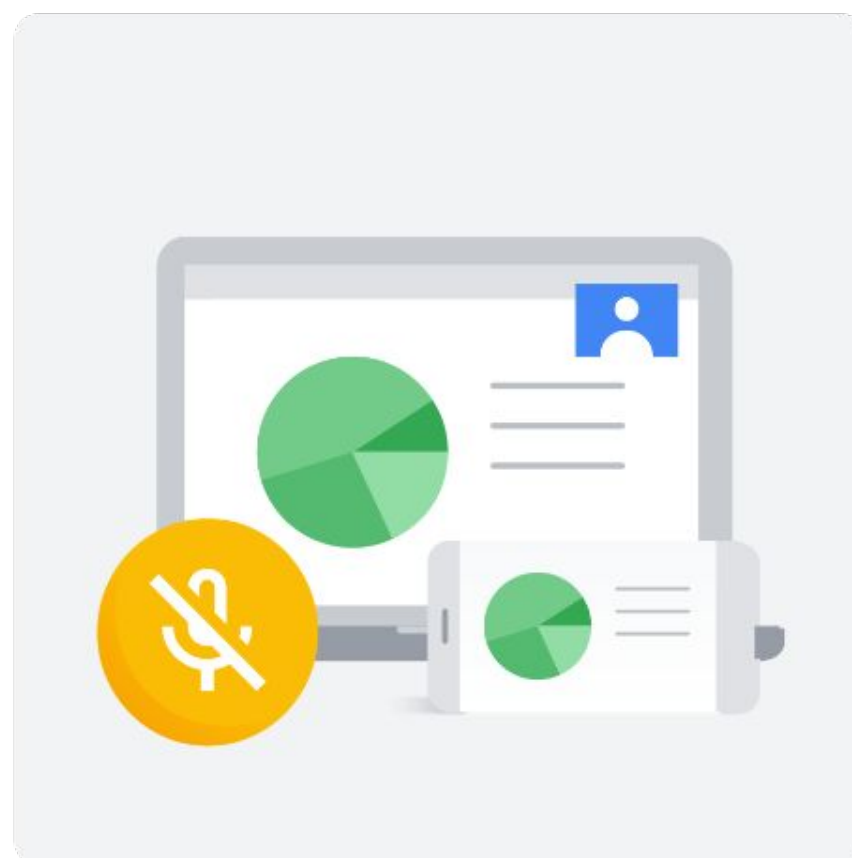
Your instructor — Imran Ahmad, PhD

Google Cloud Authorized Trainer · AWS Authorized Instructor · Microsoft Certified Trainer

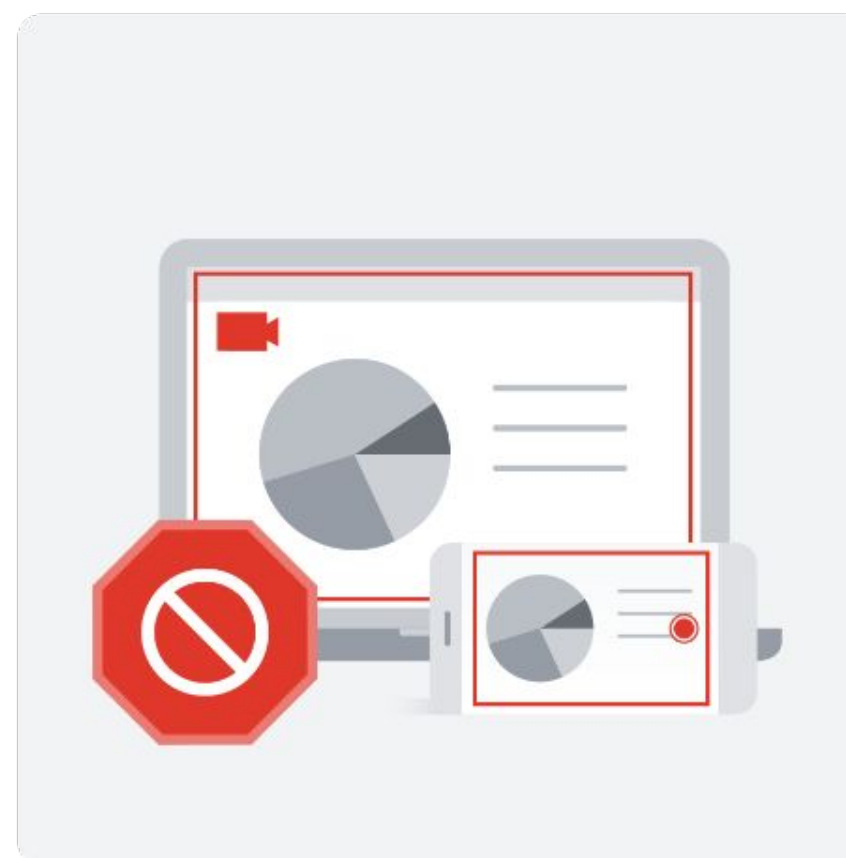
- Large-scale data engineering is the craft — production pipelines over high-volume, multi-source datasets at national scale, across Google Cloud, AWS, and Azure
- PhD in data processing — created a large-scale data-processing algorithm on cloud/grid computing; pioneered techniques applied to data-intensive science, including neutrino-particle experiments
- Taught data engineering and cloud for Citibank, JP Morgan, the United Nations, Deloitte, and the European Commission
- Second act: AI and agents — author of 30 Agents Every AI Engineer Must Build and 50 Algorithms Every Programmer Should Know (Packt)
- Fluent in your world and your target — SQL Server and SSIS to BigQuery; these two days map that craft onto your migration



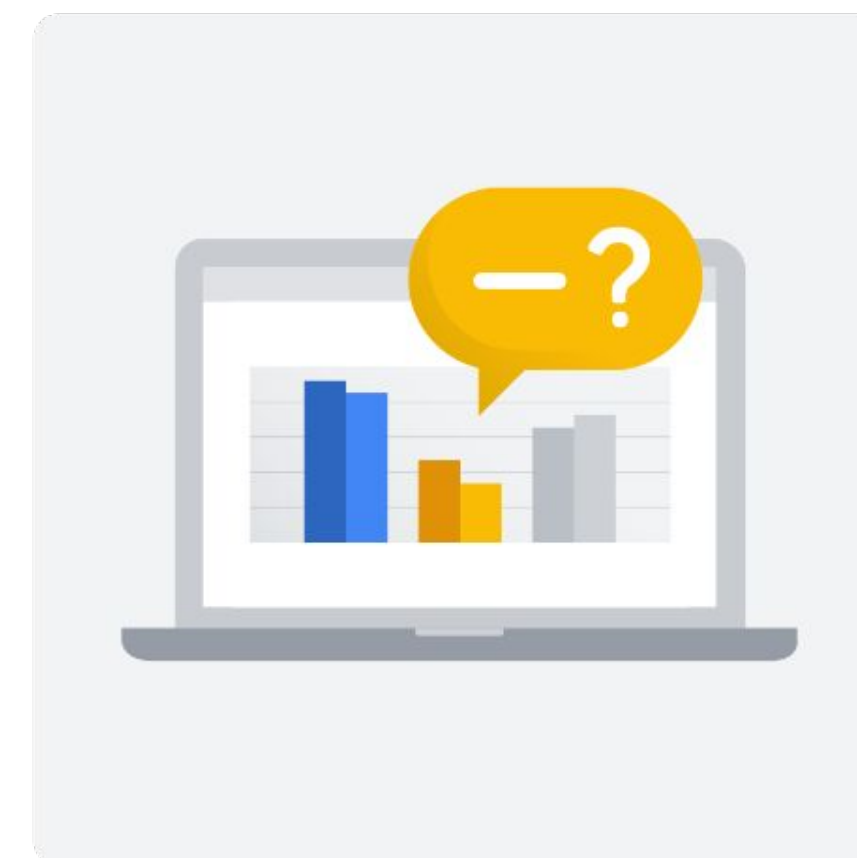
Etiquette



Mute microphone



No recording



Ask questions

The reframe: two days = your migration story

Not a generic Google Cloud tour — every module hangs on the project you are actually running.

- Kill the nightly extracts— Datastream change data capture replicates continuously, no extract window
- Land in BigQuery— serverless, no instance, no patching; your SQL Server DW's replacement
- Bronze / Silver / Gold— the medallion lakehouse from your own architecture doc, taught on Day 2
- Dataform for ELT— SQL workflows and release configs; your SSIS-in-git CI/CD discipline translates
- Power BI last mile— Gold marts feed the reports you already have (Day 2 afternoon)
- Single source of truth— BigQuery becomes the analytics source; SQL Server stays system of record

Every demo, DIN activity and lab in the next two days hangs on one of these steps.

Hands up — a quick survey before we start

Your answers set the depth for the next two days. Calibration, not a quiz.

- ETL developers— who builds and maintains the SQL Server / SSIS packages today?
- M&O team— who keeps it running and will inherit whatever we design here?
- Power BI authors— who builds the reports sitting on top of the marts?
- Who has opened the Google Cloud console— even once, even just to click around?
- Who has written a query in BigQuery, anywhere, ever?

Answer honestly, including 'never opened it' — the room is senior in DE, new to this platform.

Day 1 — Landing your data in BigQuery

09:00–09:25

Welcome, hands-up survey, the reframe (this block)

09:25–10:25

M1 DE tasks & components + console orientation — create your class dataset

DIN

10:40–11:10

Lab: Loading Data into BigQuery — your first win in a new console

LAB

11:10–13:15

M2 killing the nightly extracts — Datastream CDC lab runs across lunch

LAB

13:15–14:30

M3 extract & load + four loading DINs — autodetect pitfalls, CSV vs Parquet

DIN

14:45–16:30

M4 ELT + Dataform demo and anchor lab; M5/M6 compressed; day close

LAB

Day 2 — Lakehouse & medallion, your target model

09:00–09:15

Recap quiz — which GCP tool replaces which piece of your current stack

QUIZ

09:15–10:30

M7 lake vs warehouse vs lakehouse + M8 building a data lakehouse

10:45–12:00

Federated query lab, external tables (Bronze made tangible), M9 modernizing DWs

LAB

13:00–14:35

Iceberg lab + deep-dive; M10 governance + Google's own Bronze/Silver/Gold slides

LAB

14:50–15:20

M11 compressed — BQML direction + the Power BI to BigQuery last mile

DEMO

15:20–16:30

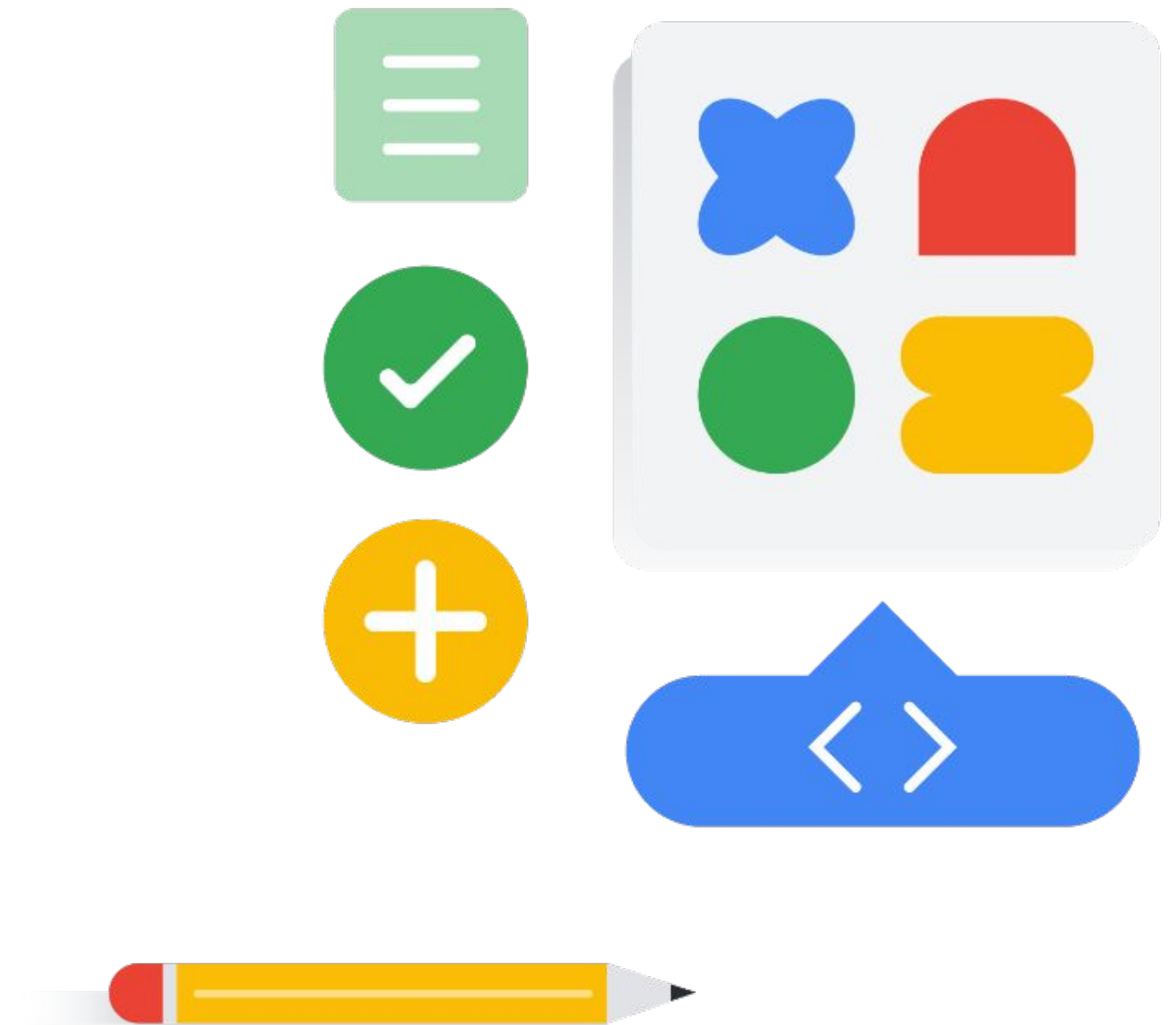
Capstone workshop: map one of your real extract jobs; summary and Q&A

WORKSHOP

Lab environment

For each lab, Google Skills offers:

- ✓ A free set of resources for a fixed amount of time
- ✓ A clean environment with permissions



Open Google Skills

- 1 Open an incognito window (or private and/or anonymous window).
- 2 Go to the URL your instructor provides.
- 3 Sign In with existing account or Join with new account (with email you used to register for the course).
- 4 Launch the course from the Dashboard page.

Access issues

The process to open Google Skills can differ based on credentials used. Please reach out to your instructor if you have any access issues.

Up next

Activities

Classrooms

SCHEDULED COURSE

Introduction to Data
Engineering on Google Cloud

▶ Join

View your labs

Do **NOT** launch a lab until instructed to do so!

Labs	Lecture Notes
	
	
	Lab Currently Disabled
	Lab Currently Disabled

- ← Lab completed
- ← To be completed
- ← Not yet available

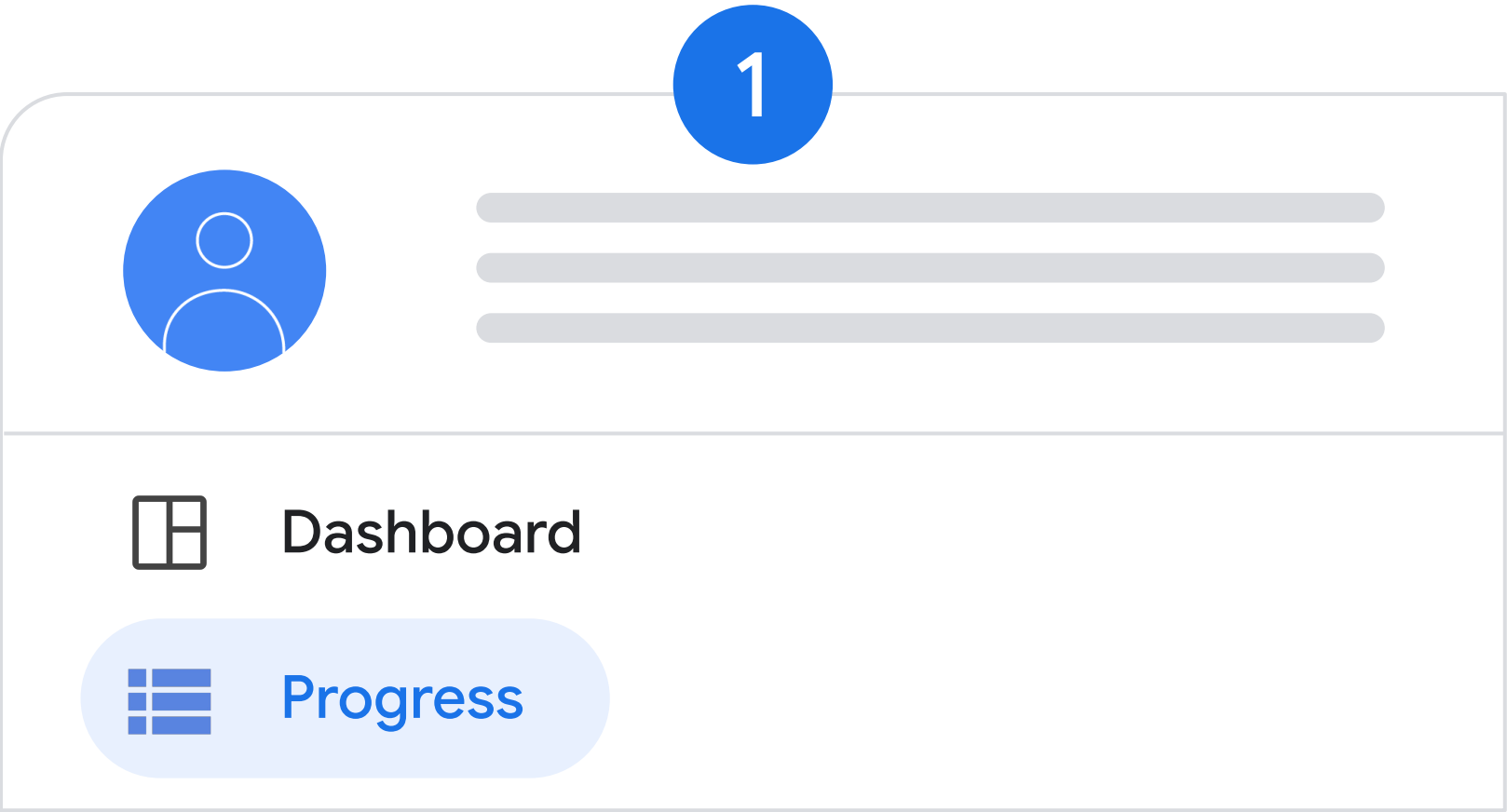
View lecture notes

Labs	Lecture Notes
01	⬇
02	⬇
03	⬇
04	⬇

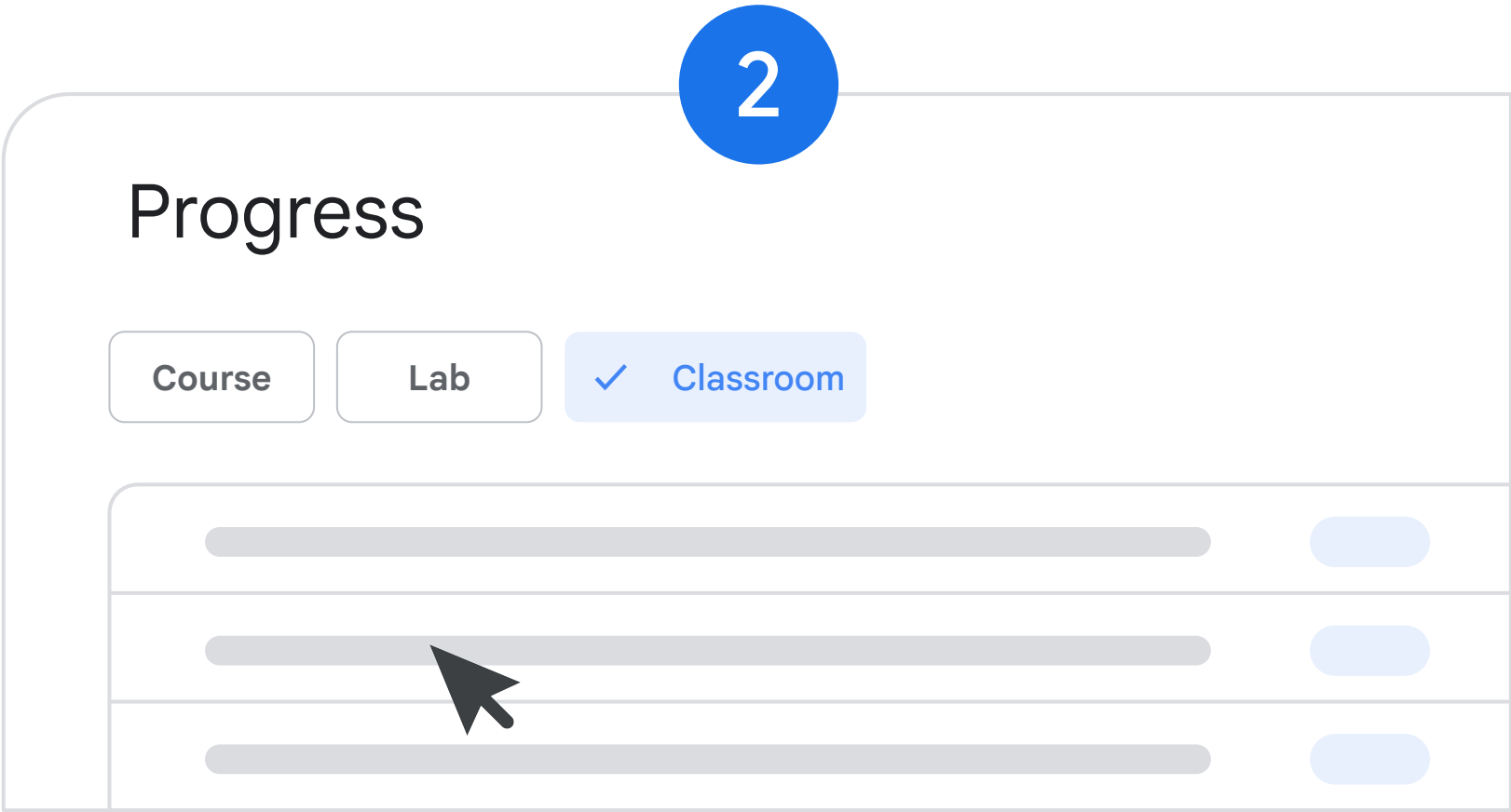
You can download these as PDF files

End of class - Materials

Materials are available for 2 years



Click on your **Profile** and select **Progress**.



Select the class from the **Classroom** list.

Logistics and ground rules

- Google Skills classroom—
<https://www.skills.google/ilt/classrooms/37990>
— all lab steps live there
- Temporary lab projects— each lab gives a throwaway GCP project plus a lab timer; nothing bills to you
- Labs are safe to break— disposable projects exist for experiments; break things freely
- Lunch 12:00–1:00 PM CT— on Day 1 the Datastream lab keeps replicating while we eat
- Breaks— 15 minutes mid-morning and mid-afternoon, both days
- Parking lot— off-track questions get written down and answered at breaks or Day 2 flex time
- Pre-course survey— fill your row if you have not yet: role, 0–3 self-ratings, one extract job you own (link in the chat and the student pack)

Keep two browser tabs ready each morning: the Skills classroom and the Google Cloud console.
Pre-course survey: survey.anum.cloud — student pack: github.anum.cloud

Take these with you — scan any code

Everything from these two days, in your pocket before you leave the room.



Pre-course survey

survey.anum.cloud



Student pack — notes, slides, labs

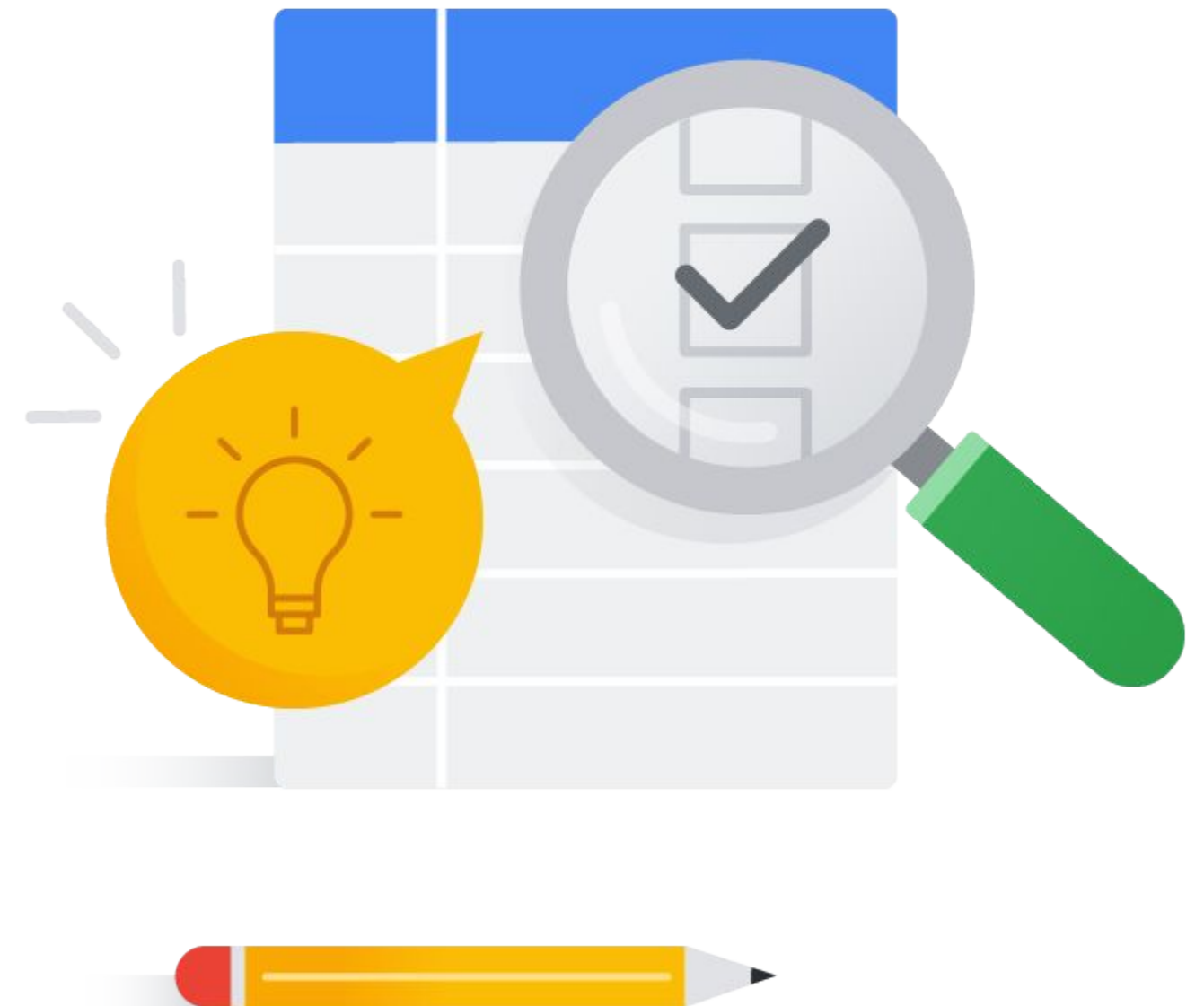
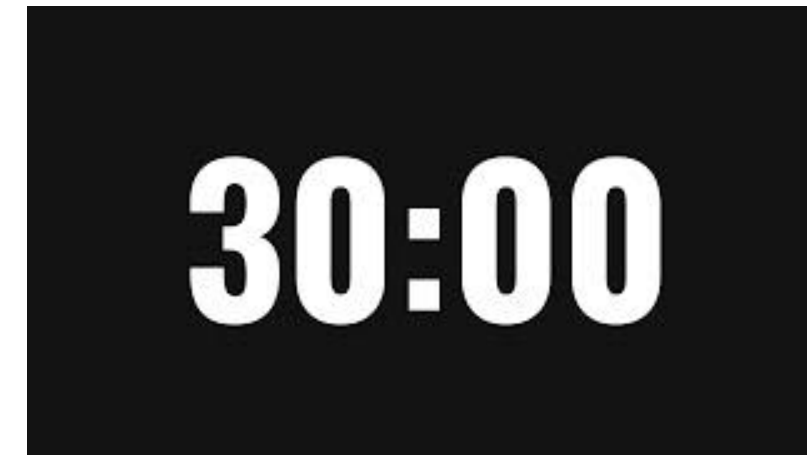
github.anum.cloud

Lab: Loading Data into BigQuery

🕒 30 min

Learning objectives

- Load data into BigQuery from various sources.
- Load data into BigQuery using the CLI and the Google Cloud console.
- Use DDL to create tables.



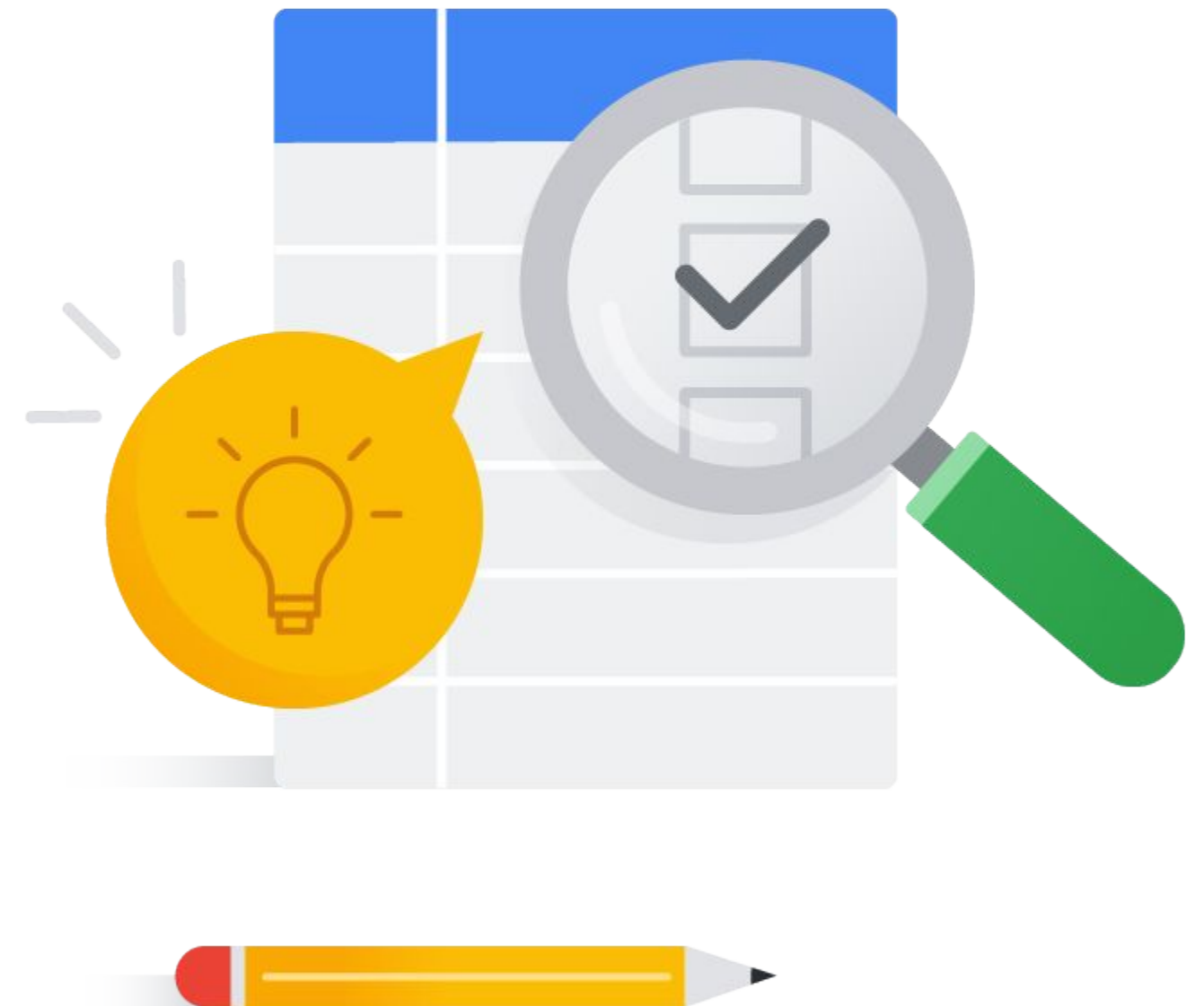
Lab (optional): Datastream: PostgreSQL Replication to BigQuery

45:00

🕒 45 min

Learning objectives

- Prepare a Cloud SQL for PostgreSQL instance using the Google Cloud console.
- Import data into the Cloud SQL instance.
- Create a Datastream connection profile for the PostgreSQL database.
- Create a Datastream connection profile for the BigQuery destination.
- Create a Datastream stream and start replication.
- Validate that the existing data and changes are replicated correctly into BigQuery.

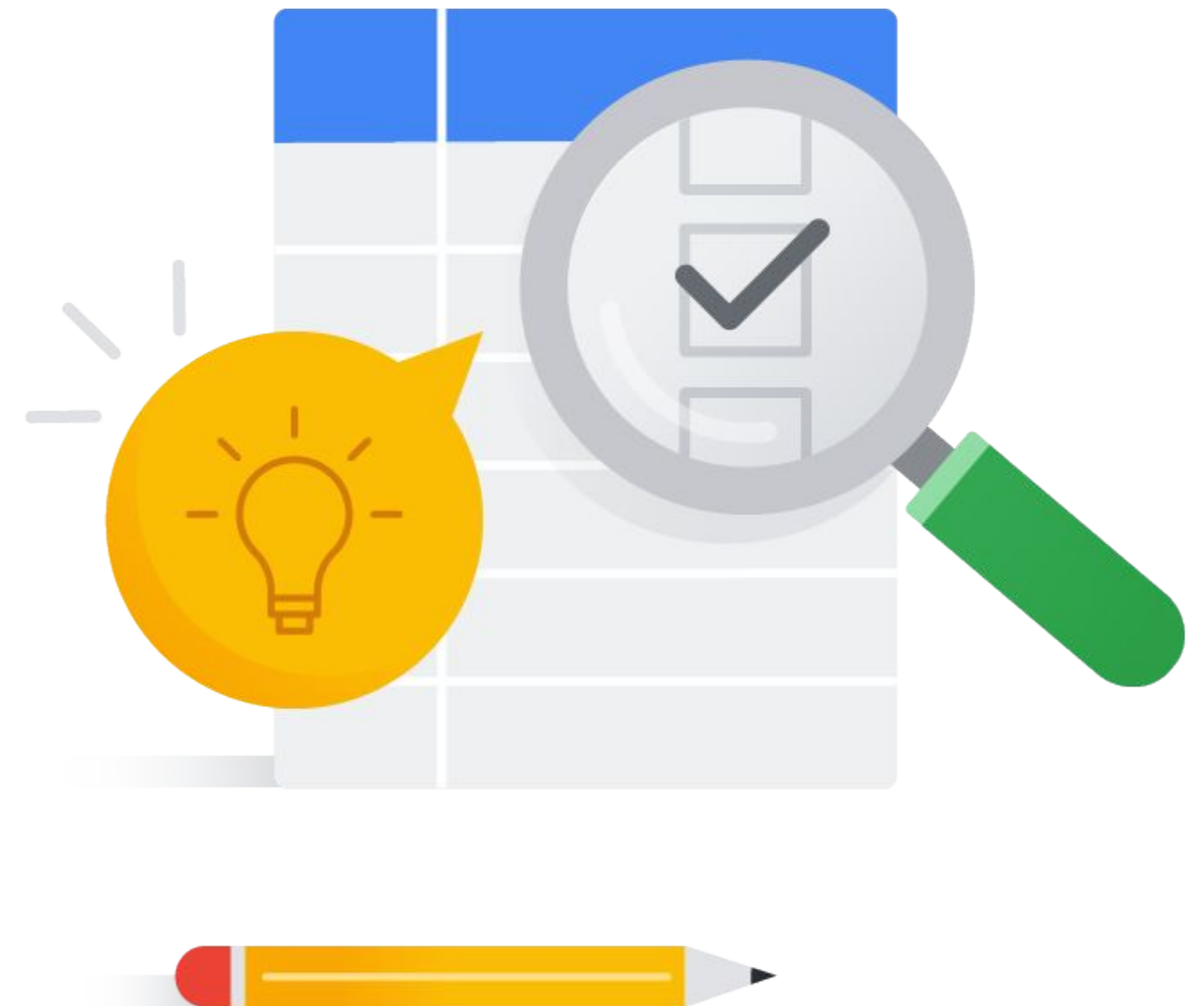
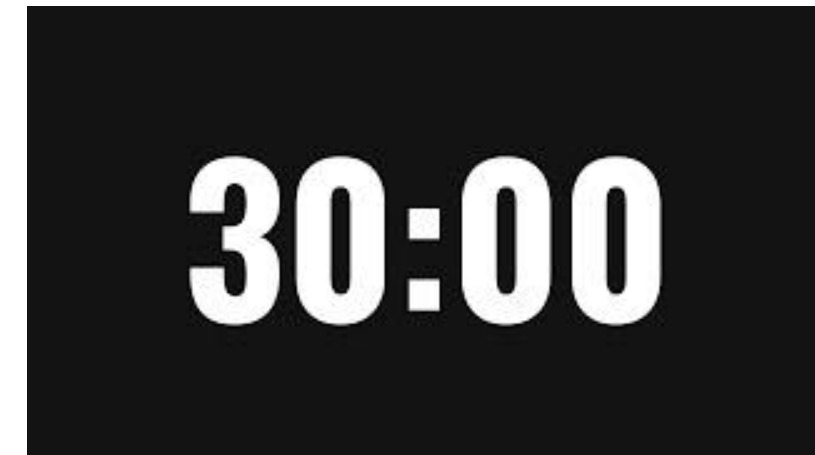


Lab: Create and Execute a SQL Workflow in Dataform

🕒 30 min

Learning objectives

- Create a Dataform repository.
- Create and initialize a Dataform development workspace.
- Create and execute a SQL workflow.
- View execution logs in Dataform.



Google Cloud

Next at 09:25 — M1: Data Engineering Tasks & Components

- ✓ You have the shape of the two days— kill the extracts, land in BigQuery, medallion, Dataform, Power BI
- ✓ M1 maps your SSIS mental model 1:1— sources, transforms, sinks with Google service names on each box
- ✓ And orients you in the platform— console, projects, IAM, billing, regions
- ✓ At 10:10 you go hands-on— star two projects and create your class dataset (everything after lunch needs it)
- ✓ Tabs ready: Skills classroom + Google Cloud console